AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning

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Project Proposal Report

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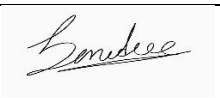
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Orchid cultivation plays a significant role in the ornamental plant industry, specifically in tropical regions, where Vanda orchids are highly valued for their bright flowers and ability to bloom several times throughout the year [2]. However, identifying the growth stages of Vanda orchids and predicting the flowering time is challenging due to their unique biological features, such as their one-sided growth, aerial roots, and irregular flowering cycles[3]. Currently, most farmers rely on physical monitoring and experience to observe plant growth and flowering patterns. These traditional methods are time consuming and difficult to apply effectively in large scale orchid farms.

Recent research has introduced high level monitoring systems using multi modal sensing and machine learning to monitor orchid development conditions. However, most existing studies specifically focus on species such as Phalaenopsis and focus on general plant health monitoring rather than detailed growth stage classification and flowering cycle prediction. Furthermore, limited focus has been given to modeling entire flowering life cycle of Vanda orchids, including the post blooming stages. This highlights the research gap in developing intelligent systems capable of identifying specific growth patterns of Vanda and supporting full life cycle monitoring [6].

This research proposes an AI based model for orchid growth stage classification, flowering prediction, and post flowering monitoring for Vanda species as a component of an AI powered smart orchid care system. The system will use computer vision and machine learning techniques to analyze orchid images and environmental data such as temperature and humidity. Deep learning models and image processing techniques predict future flowering times and classify growth stages, including seedling, vegetative growth, flowering, and post flowering stages.

The expected result is a smart decision support system that helps farmers improve cultivation practices, plan harvest schedules, and enhance breeding management. This solution can improve productivity, reduce crop losses, and increase profitability in the ornamental flower industry while supporting sustainable orchid cultivation.

Keywords: Orchid Growth Stage Classification, Bloom Prediction, Post-Bloom Monitoring, Vanda Orchids, Multi-Modal Machine Learning.

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LIST OF ABBREVIATIONS

SE -Software Engineering

SLIIT - Sri Lanka Institute of Information Technology

AI - artificial intelligence

SDG - Sustainable Development Goal

NDVI - Normalized Difference Vegetation Index

PRI - Photochemical Reflectance Index

WBI - Water Band Index

WBS - work breakdown structures

CNN - Convolutional Neural Network

LSTM - Long Short-Term Memory

MAE - Mean Absolute Error

GPL - General Public License

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1. INTRODUCTION

1.1 Background and Context

Orchid cultivation is an important agricultural activity throughout Asia, and the Vanda orchid is highly valued for its ability to bloom many times throughout the year and is sold in export markets. However, farmers face critical problems in managing their crops efficiently. Studies show that 12-18 percent of production is lost due to poor environmental management. [1]. Vanda orchids have unique characteristics. They grow tall on a single stem, have aerial roots, and can bloom at any time. These characteristics make it very difficult to monitor their growth using traditional methods [3].

Small and medium-sized farmers are the ones who suffer the most from these difficult conditions. They cannot pay for advanced technology and must rely on workers to visually inspect the plants. This manual method is slow, inconsistent and often too late. When farmers cannot predict when their orchids will bloom, they fail to sell or harvest the flowers at the best time. It can be too early or too late. This reduces both the quality of the flowers and the price they can get.

Research shows that plant stress is only detected by visual inspection by workers 5-7 days after it begins to occur; by this time, a reduction in yield has already occurred [1]. Other research has shown that precise monitoring technology can lead to a 15-30 % reduction in resource use in floriculture [5]. These statistics show that existing methods are failing and that new technology can be a great help.

If we analyze the existing solutions, we can see that there are some problems. For example, a multi-modal sensor system is well described in the literature [1]. This system is good for Phalaenopsis orchids, as it is able to detect nutrient problems and stress with an accuracy of 94.2% 72 hours before visible signs become apparent. However, this solution is not intended for Vanda orchids [1]. The fact that Vanda orchids bloom constantly, that these orchids have a different structure, and how these orchids behave after flowering are not described in this research. In this research, an accuracy of 89.4% was achieved in predicting flowering 21 days in advance, but this solution has not been made available to farmers and has not been tested on Vanda orchids [1].

The proposed solution is an artificial intelligence (AI) based model that can identify the growth stages of Vanda orchids and predict flowering. It can use images and data from sensors to identify different growth stages, such as young seedlings, leaf and root growth, flower bud development, full bloom, and finally recovery after flowering. By learning how these plants grow, it can:

- Predict when they will bloom so that farmers can prepare for harvesting
- Predict how long they will bloom so that distribution can be arranged
- Predict when seeds can be collected after pollination so that they can be propagated

Unlike traditional models, an artificial intelligence-based model will continue to learn and improve. It can account for hybrids like Vanda and changes in weather.

This research contributes to SDG 12 (Responsible Consumption and Production), helping farmers conserve resources by using efficient time management during harvest and minimizing post-harvest waste. This research also contributes to SDG 9 (Industry, Innovation and Infrastructure) by introducing artificial intelligence and machine learning to farming.

Research Cluster Alignment

This research is part of a research cluster on the applications of artificial intelligence in precision agriculture and smart farming. This research contributes to the automated monitoring of plant growth, computer vision in agriculture, and machine learning for crop growth prediction.

1.2 Literature Review

Recent studies in orchid monitoring have been based on the application of optical sensing techniques to monitor plant health and growth. A study by Sungehta et al. [1] proposed a multi-modal sensor system for orchid cultivation that included hyperspectral imaging, fluorescence spectroscopy, and machine vision. The results of the study were promising, with the system achieving an accuracy rate of 94.2% for nutrient detection, stress detection 72 hours before symptoms, and an accuracy rate of 89.4% for bloom prediction 21 days before. The study showed that a combination of sensor technologies reduced the false positive rate by 60% compared to single sensor technologies. However, the study was based on Phalaenopsis orchids and did not take into account the growth characteristics of Vanda orchids, especially with regard to continuous blooming and recovery phases.

Another study by Srisompun et al. [4] proposed a classification method for Paphiopedilum orchids using image processing and neural networks. In this study, the researchers developed a dataset of 1,100 images including 11 orchid species. The study combined color instances, color histograms, and texture features, and fed these features to a multilayer perceptron neural network. The study reported an accuracy rate of 97.64 percent in classified orchids. Although high accuracy was reported in classifying orchid species, only orchid species were classified, and growth stages and flowering times were not determined.

A study by Chen et al. [5] proposed a self-supervised learning method when examining orchids in a greenhouse environment. This study applied ResNet-50 for feature extraction and fine-grained classification to identify 219 orchid species. The study reported an increase in accuracy from 90.9% to 98.6% through data enhancement. The study also used IoT sensors to monitor environmental factors. Although high accuracy was reported in identifying orchid species, growth stages and flowering times were not determined.

The limitations of the existing systems can be listed as follows:

- Research is conducted on species identification [4, 5] or stress detection and resource management [1] but not on growth stage classification and bloom prediction for Vanda orchids.
- Although the systems for classification are very accurate, they cannot be used to identify the growth stage classification of plants.
- The system for bloom prediction, although very accurate, is based on research in [1], and its accuracy is not for Vanda but for Phalaenopsis.
- None of the systems takes into account the unique continuous flowering nature of Vanda orchids.
- The recovery period after flowering is completely ignored in the research.
- The systems also use expensive hardware such as hyperspectral cameras for this purpose, which is not affordable for small farmers.

These gaps justify the need for a specific Vanda orchid growth stage classification and flowering prediction system that is affordable, easy to use, and covers the entire growth cycle, including recovery after flowering.

1.3 Research Gap

After reviewing the existing research in the area, it can be determined that there are a number of clear gaps in the research. The existing research focuses on three areas: orchid species identification, stress detection, and environmental monitoring. Chen et al. [5] have shown a high accuracy of 98.6 percent in their research in identifying orchid species but cannot monitor plant growth. Srisompun et al. [4] have shown a high accuracy of 97.64% in classifying Paphiopedilum orchid species, but only in species identification. Sungehetha et al. [1] have shown a high accuracy of 89.4% in predicting flowering in Phalaenopsis orchids, not Vanda species.

The methods currently used are deficient for several reasons. First, there is a lack of a system specifically designed for Vanda orchids and the continuous nature of their flowers. Second, the classification of growth stages, i.e., seedling, vegetative growth, pod initiation, flowering and post-flowering recovery, is completely absent in all studies. Third, the post-flowering phase is completely ignored, which is an integral part of preparing the plants for the next cycle. Fourth, these methods require expensive equipment such as hyperspectral cameras [1], which small farmers cannot afford.

The novelty of our research is that we propose the first growth stage classification and flowering prediction system specifically tailored for Vanda, covering the entire growth process, including post-flowering recovery. This is an important research gap and a real need in practical farming.

Table 1.1: Research Gap Comparison

Research Paper	**What it does**	**Research Gap**	**Features & Technology Used**
Multi-Modal Optical Sensing Fusion for Orchid Monitoring (2025)	Integrates hyperspectral imaging, fluorescence spectroscopy, and machine vision for real-time monitoring of orchid growth parameters (photosynthetic efficiency, leaf area index, early stress detection) in	Focuses primarily on Phalaenopsis species; does not address Vanda-specific growth habits (monopodial, aerial roots, year-round flowering); lacks models for continuous flowering prediction; no post-bloom recovery monitoring	Hyperspectral Imaging (NDVI, PRI, WBI indices; 72h early stress detection; 94.2% nutrient accuracy) Fluorescence Spectroscopy (Fv/Fm ratio; Fv/Fm >0.8 = 43.2% more flowers) Machine Vision (±0.27mm/day spike tracking; 98% accuracy)

	commercial cultivation		Edge/Fog/Cloud Computing (1.87s processing; 37.2% better accuracy) Sensor Fusion (60% fewer false positives) Machine Learning (89.4% bloom prediction at 21-day lead time)

2. OBJECTIVE

2.1 Main Objective

Using data from cameras and sensors, we designed and tested an AI-assisted system to classify and predict growth stages and flowering events for Vanda orchids with at least 90 percent accuracy in identifying critical growth stages, such as post-flowering recovery, and predicting flowering events up to 21 days in advance.

2.2 Specific Objectives

1. To develop a labeled image dataset for Vanda orchids at different growth stages such as seedlings, vegetative growth stage, pod formation, flowering stage and post-flowering recovery stage under different conditions in a greenhouse environment, with environmental sensor data (temperature, humidity).
2. To create a machine learning model that can predict and classify different growth stages of Vanda orchids using images. Since this stage is neglected in existing research, special attention should be paid to accurately classify the post-flowering recovery stage.
3. To create a bloom prediction model that can analyze data from growth stages and environmental conditions to predict orchids bloom 21 days before actual bloom with a high accuracy rate of at least 80 percent, allowing farmers to prepare for orchid harvest.
4. Integrating a camera and small environmental sensors (temperature, humidity) into a single device to collect data and transmit it to a cloud-based platform for analysis.

3. METHODOLOGY

The methodology for the proposed research is to use design science research with experimental validation. This methodology is applicable to the proposed research as a new AI system is to be developed to classify the growth stages of Vanda orchids as well as predict the flowering time.

The system will use two primary machine learning models. One model is a Convolutional Neural Network (CNN) that will be used to classify images of orchids at different growth stages. The growth stages include seedling, vegetative growth, bud initiation, flowering, and post-flowering recovery. As in other machine learning-based orchid classification studies, the CNN model will be effective in recognizing patterns in images [4, 5]. The other model is a Long Short-Term Memory (LSTM) model that will be used to predict flowering dates up to 21 days in advance by analyzing patterns over time and other environmental factors such as temperature and humidity. Research shows that such a model can achieve an accuracy rate of 89.4% in flowering prediction [1].

For all developments, Python will be the main programming language. Python is free and many AI libraries are available. TensorFlow or PyTorch will be used to develop and train deep learning models. The reason for choosing these is that they are open source and many people use them for similar research [4]. OpenCV will be used for image preprocessing tasks such as resizing and cropping. Hardware requirements include a Raspberry Pi camera for daily images of the plant. DHT22 sensors will be used for temperature and humidity sensing. The components are inexpensive and can be easily installed in the greenhouse [5]. Firebase will be used for free cloud storage. Google Colab will be used for free training of the model.

Images of Vanda orchids will be collected daily from the greenhouse for three to four months. Images will be collected at specific times each day from a specific camera location. Environmental data also will be collected from green house. Images will be appropriately labeled according to growth stage with the advice of experts in the field. The goal is to collect at least 1,000 images indicating all growth stages. Permission will be got from Greenhouse owners before data collection. No personal data will be collected from farmers. Farmers can leave at any time from the study. The data collected will be used just for this study.

The dataset is divided into separate parts. 70% for training, 15% for validation, and 15% for testing. Ten-fold cross-validation ensures that the model performs well on unobserved data.. For growth stage classification, accuracy, and F1-score measures are used, with a target of greater than 80%. For flowering, the mean absolute error (MAE) compute of the average error in days is used, with a target of less than three days. The final system will be tested in a greenhouse, comparing the predicted blooming daysof flowers with the actual blooming dates of flowers, along with farmer feedback. Research shows that similar methods for validation obtains 97.64% accuracy for orchid classification and 98.6% accuracy for orchid identification [5].

4. HIGH-LEVEL SYSTEM ARCHITECTURE

- Overview of proposed solution and its components
- System diagram of components
- Highlight the individual component and its parts
- Component interactions
- Data flow explanation
- Scalability and security considerations

The proposed system is an AI Powered Smart Orchid Care System Using Multi-Modal Machine Learning solution. it helps farmers manage Vanda orchids more successfully. Hybrid pollen analysis, disease detection, smart fertilization, and growth stage classification with bloom prediction are the 4 components of the whole system that works together to form the complete system. This report focuses on the growth stage classification and bloom prediction component for Vanda orchids and briefly mentions how it is connected to other components.

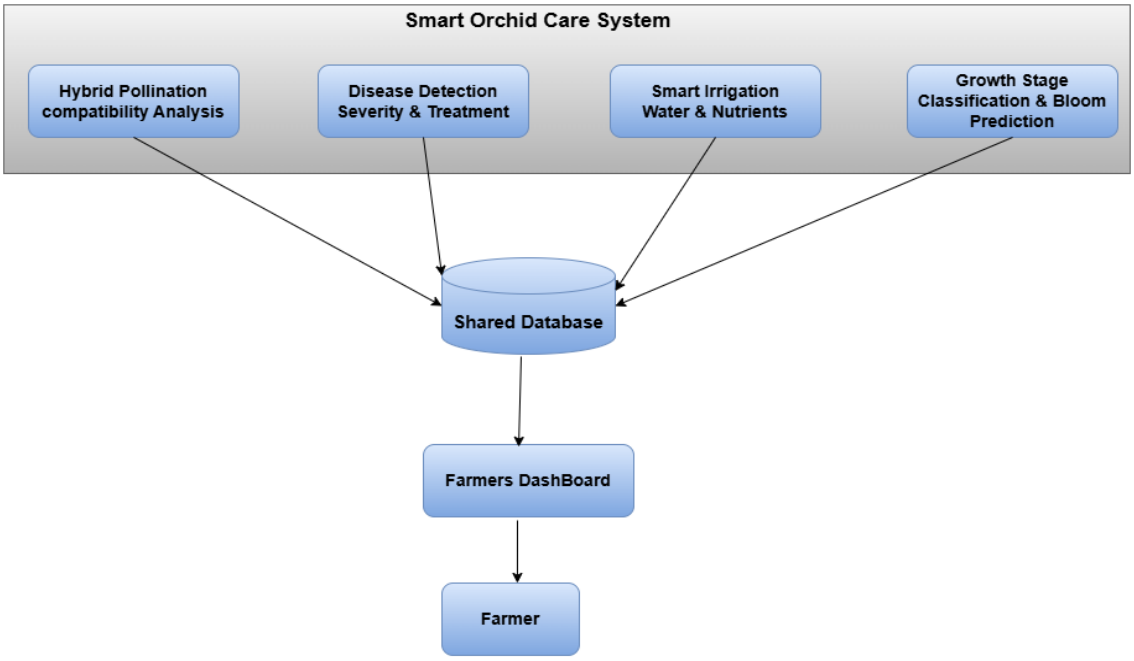


Figure 4.1: Four components of the system

Using image and environmental sensor data, the growth stage classification and bloom prediction component automatically recognize Vanda orchid growth stages and predict flowering times. This component consists of a camera module using a Raspberry Pi camera that captures images daily of every Vanda plant, and environmental sensors (DHT22) that measure temperature and humidity. A CNN classification model acts as a deep learning model that recognizes which growth stage the plant is in, while an LSTM prediction model serves as a time series model that forecast flowering dates of Vanda orchid flowers up to 21 days in the future. before syncing them to the cloud, a local database stores images, sensor data, and predictions, and an API layer link this component to the shared system database.

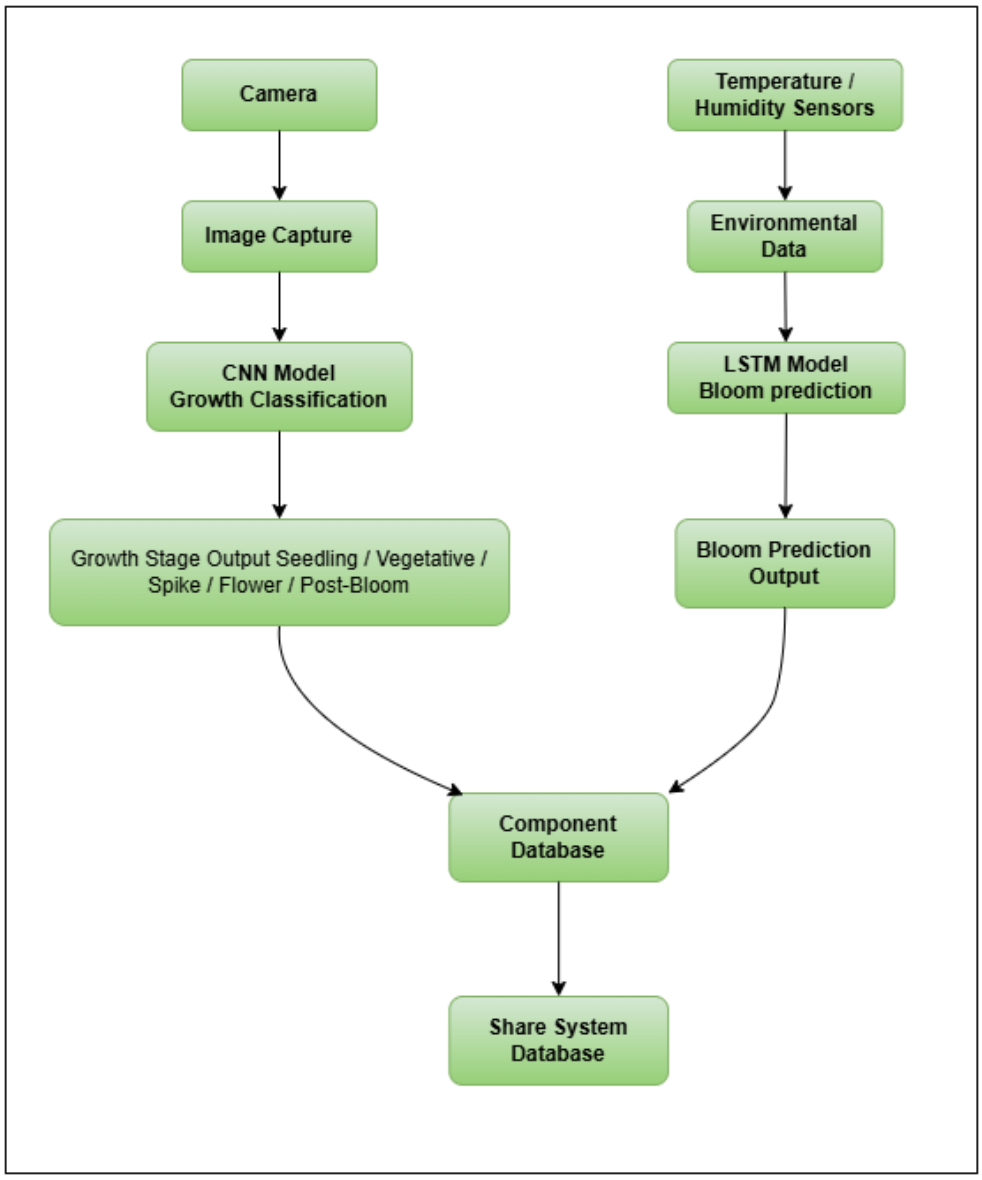


Figure 4.2: System diagram of growth stage classification and bloom prediction component

Even if each component works separately, they share data to create a complete picture for the farmer. My component provides the current growth stage to help the Pollen component recognize when the plant is ready to breed and provides the growth stage history to help the Fertilizer component determine fertilizer needs (higher during the young stage, lower during recovery after flowering). In turn, my component Growth Stage and Bloom Prediction receive disease alerts from the Disease Detection component to help explain unexpected growth patterns and receives water and nutrient schedules from the Fertilizer component, as these impacts on growth rates and blooming times of Vanda.

The data flow starts with data collection, where the camera takes single photo per plant at the same time each day, and the sensors record temperature and humidity every hour. All data is saved on a Raspberry Pi with time stamps for local storage. At the end of each day, the data is uploaded to the cloud database using the Firebase Free Tier. The CNN model then analyzes the image and recognizes

the growth stage in five categories: seedling, vegetative, pod, flower, or post-bloom. The LSTM model uses the growth stage history and environmental data to predict when flowering will occur 21 days in advance. The results are saved in the component database and then synced to the shared system database so that other components can use them. Finally, the farmer sees the growth stage and predicted bloom date of the dashboard.

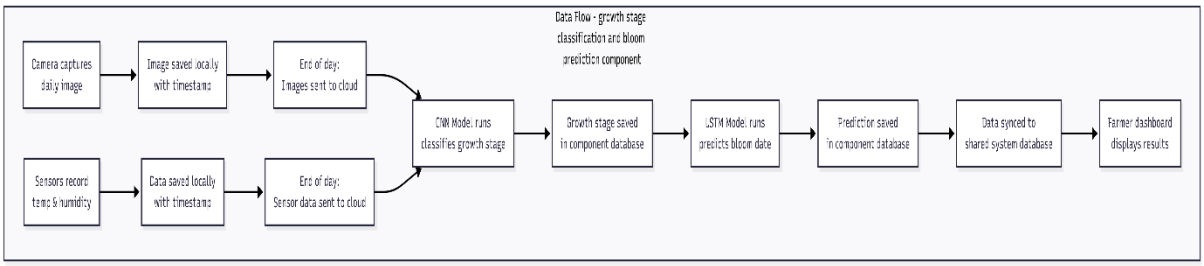


Figure 4.3: Data flow of growth stage classification and bloom prediction component

For scalability, if more plants are added, the camera can be moved on a rail system to cover 50-100 plants per unit. Each greenhouse gets its own hardware unit while the cloud handles multiple locations, if more greenhouses are added. As more data is collected over time, Firebase auto scales and the free tier is enough for the pilot phase. Role-based access is provided for the administrator, farmer, and worker levels for multiple users.

For security reasons, Farmer data is not shared with any third parties and is anonymized for research purposes. Login is required to view the dashboard with different access levels for different user levels. All data sent to the cloud is encrypted using HTTPS. The hardware units are kept in locked boxes so that only authorized people can physically access them, and data is backed up daily to prevent data loss.

5. USER REQUIREMENTS

Stakeholders Identification

The main stakeholders of this system are orchid farmers and growers who will be interested in monitoring the growth of the plants and their full bloom time. Daily monitoring of the plants by greenhouse workers is also done using the system. Orchid breeders and extension officers can also use the system. In addition, system administrators are responsible for the operation of the system.

Functional requirements

1. The system should capture images of Vanda orchids at regular intervals at least once a day via fixed cameras in the greenhouse.
2. The system should collect data from environmental sensors, including temperature and humidity, once an hour.
3. The system should analyze the images and identify the growth stage of the orchids, classifying them into one of five growth stages: seedling, vegetative growth, pod initiation, flowering, and post-flowering recovery.
4. The system should analyze growth patterns and environmental data to predict the flowering time of an orchid at least 21 days in advance.
5. The system should display information about the current growth stage and predicted flowering date for each orchid on a user-friendly dashboard via a web browser or mobile app.
6. The system should notify farmers when an orchid is expected to flower in the next 7 days.
7. The system should store historical growth data for each orchid, enabling analysis and improvement in predicting growth stages and dates.
8. The system also allows users to manually update incorrect growth stages, allowing users to provide feedback and improve model accuracy.

Non-functional requirements

- Performance: The system will be able to process images and provide a classification of growth within 5 seconds. Updates to flowering will be made within 30 minutes after receiving fresh data.
- Usability: The interface will be simple and easy to understand, with minimal technical jargon. Icons and local language support will be provided as needed for farmers with less technical knowledge. Since most farmers use them in greenhouses, the interface can be accessed from mobile phones.
- Security: Password protection will be used for user accounts. Different users such as administrator, farmer and worker will have different access levels. All data exchanged between sensors; cameras and the cloud will be encrypted. No farmer data will be shared with any third party.
- Scalability: The system architecture will be such that additional cameras and sensors can be added as the farm grows. New orchid hybrids will be incorporated into the system without the need to retrain the entire model.

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

The target audience is Asian orchid growers involved in the Vanda orchid cultivation business. Vanda orchids take three or more years to reach the flowering stage. Therefore, monitoring these orchids is very important to ensure efficient growth. The orchid industry in the Asia Pacific region has Vanda orchids as one of its main production sectors. Some of the major orchid growing countries are Thailand, Taiwan, and China [7].

The target customer is a small or medium-sized orchid grower with between 500 and 5,000 Vanda orchids grown in a greenhouse. These growers currently rely on manual inspection by employees to monitor whether the orchids are blooming or not. They are currently losing revenue due to the unstable flowering time. According to research reports, the orchid industry is experiencing a growth rate of 10 - 15% every year. This is a great business opportunity for growers who can accurately monitor flowers to maximize revenue [8]. Secondary target audiences are orchid breeders and agricultural extension services.

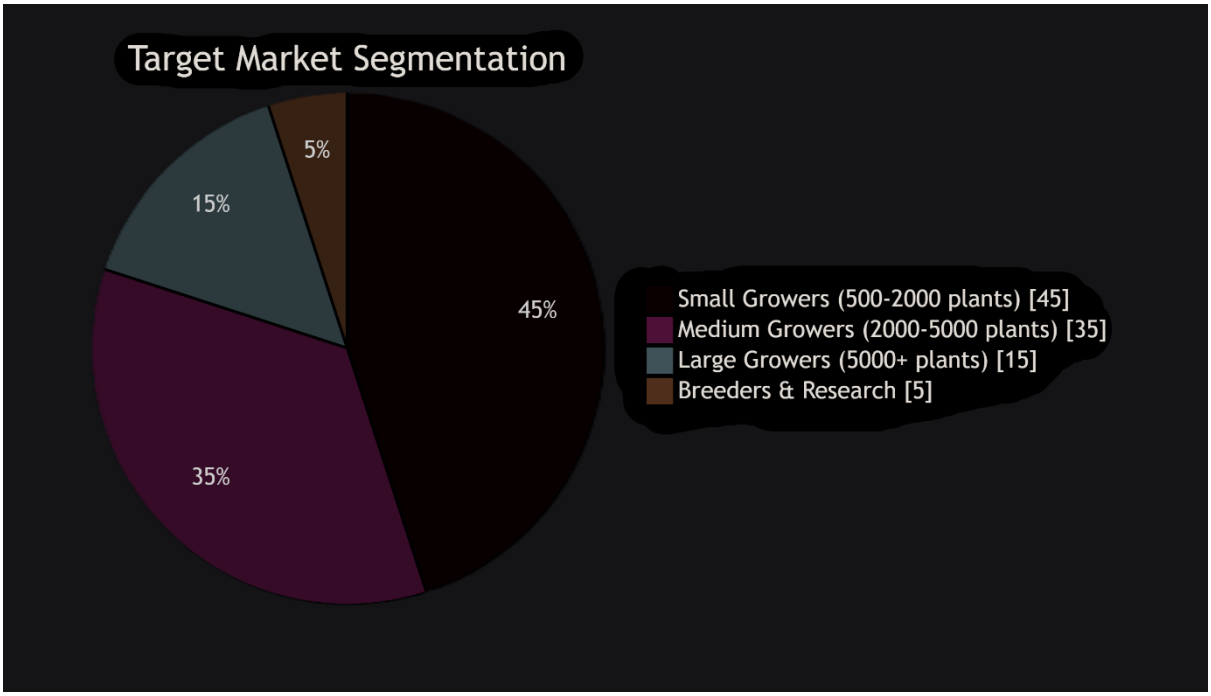


Figure 6.1: Target Market Segmentation

Value proposition

The system will enable accurate identification of orchid growth stages and prediction of bloom time, especially for Vanda orchids. This will be particularly important due to the continuous nature of orchid flowering. Farmers will be notified 21 days before orchids are ready to be picked. This will enable them to plan for picking and marketing.

Research has shown that accurate monitoring can help reduce production losses of between 12 and 18 percent due to poor environmental management. Accurate monitoring will also help improve farmers' incomes due to reduced marketability and quality of orchids. A study on the

profitability of orchid cultivation showed that optimizing production decisions improves farmers' incomes.

Revenue model

The revenue model is based on a software-as-a-service approach with different subscription plans:

- **Basic Plan:** Rs. 1,500 - 2,500 per month, up to 500 plants, growth stage classification
- **Professional Plan:** Rs. 3,500 - 5,000 per month, 500 - 2,000 plants, flowering prediction
- **Enterprise Plan:** Rs. 6,000 - 8,000 per month, 2,000+ plants, post-flowering tracking, multi-user support

Revenue is also generated from:

- **One-time installation and training costs:** Rs. 10,000 - 15,000
- **Hardware Packages:** Rs. 15,000 - 25,000

Cost estimation

Table 6.1: Cost Estimation

Item	Amount (LKR)
One-Time Cost (to build the product)	
App development	20,000
Testing with farmers	10,000
Total One-Time	**30,000**

Cost Per Farmer (to set up one customer)
Hardware (camera, sensors, Raspberry Pi) - LKR 15,000
Installation and travel – LKR 2,000
Total Per Farmer - LKR 17,000

Pricing strategy

Table 6.2: Pricing strategy

Plan	**Monthly Price**	**What Farmer Gets**
Basic	LKR 500	Growth stage only
Premium	LKR 1,000	Growth stage + bloom prediction + post-bloom alerts

Competitive advantage

Table 6.3: Competitive advantage

Competitor Type	**Limitations**	**Our Advantage**
General crop monitoring systems	Not designed for orchids, no Vanda features	Species-specific for Vanda
Research systems	Expensive hyperspectral sensors	Affordable cameras
Manual visual inspection	Slow, inconsistent, 5–7-day delay	Automated, 21-day prediction
Smart commercial farming systems	Very expensive, LKR 100,000+ [14]	Budget-friendly, LKR 15,000

Intellectual property considerations

The core intellectual property consists of trained machine learning models for Vanda growth stage classification and bloom prediction. Copyright protection is required for the software code. The methodology for integrating affordable sensors with AI for Vanda specific prediction is protected as a trade secret. Since open-source tools are used, the code can be protected under relevant open-source licenses (such as the GNU GPL), which enables use while protecting the core algorithms.

7. BUDGET AND JUSTIFICATION

Table .7.1: Budget Plan

Hardware Costs (per unit)			
Cost Category	**Estimated Amount (LKR)**	**Notes**	**Justification**
Raspberry Pi Camera Module	4,500–6,000	Captures plant images	Captures daily images of Vanda orchids to track growth stages (seedling, vegetative, spike, flower, post-bloom).
DHT22 Temperature/Humidity Sensor	900–1,200	Measures environment	Measures greenhouse environment. Temperature and humidity are key factors affecting flowering time and must be included in the prediction model.
ESP32 Microcontroller	2,000–3,000	Processes sensor data	Reads sensor data and sends it to the cloud via WiFi. Low cost and low power consumption.
Raspberry Pi Zero 2W	3,500–4,500	Runs image processing	Runs the image processing at the edge. It captures and analyzes images before sending results to the cloud.
Power supply, cables, SD card	2,500–3,500	Basic components	Provides stable power to the Raspberry Pi and sensors. Stores the operating system and temporary image data. Connects sensors to the board.
3D printed enclosure	1,000–1,500	Protects electronics	Protects electronics from greenhouse humidity, dust, and heat.
Total Hardware per Unit	**14,400–19,700**		
Software Costs			
Cost Category	**Estimated Amount (LKR)**	**Notes**	**Justification**
All software (Python, TensorFlow, etc.)	0	Open source	Free, open-source
Total Software	**0**		
Cloud Services (monthly)			
Cost Category	**Estimated Amount (LKR)**	**Notes**	**Justification**
Firebase free tier	0	Up to 1 GB storage	Provides database and storage for free up to 1GB, which is enough for a pilot project with one unit.
Total Cloud (Year 1)	**0**		
Human Effort Costs			
Cost Category	**Estimated Amount (LKR)**	**Notes**	**Justification**
Travel to greenhouses (fuel, bus fare)	5,000–8,000	10–15 visits	
Mobile data/Wi-Fi charges	5,000–8,000	For cloud testing and updates	

Refreshments during fieldwork	4,000–6,000	Snacks/meals during long days	
Total Human Effort	**14,000–22,000**		

8. WORK BREAKDOWN STRUCTURE (WBS)

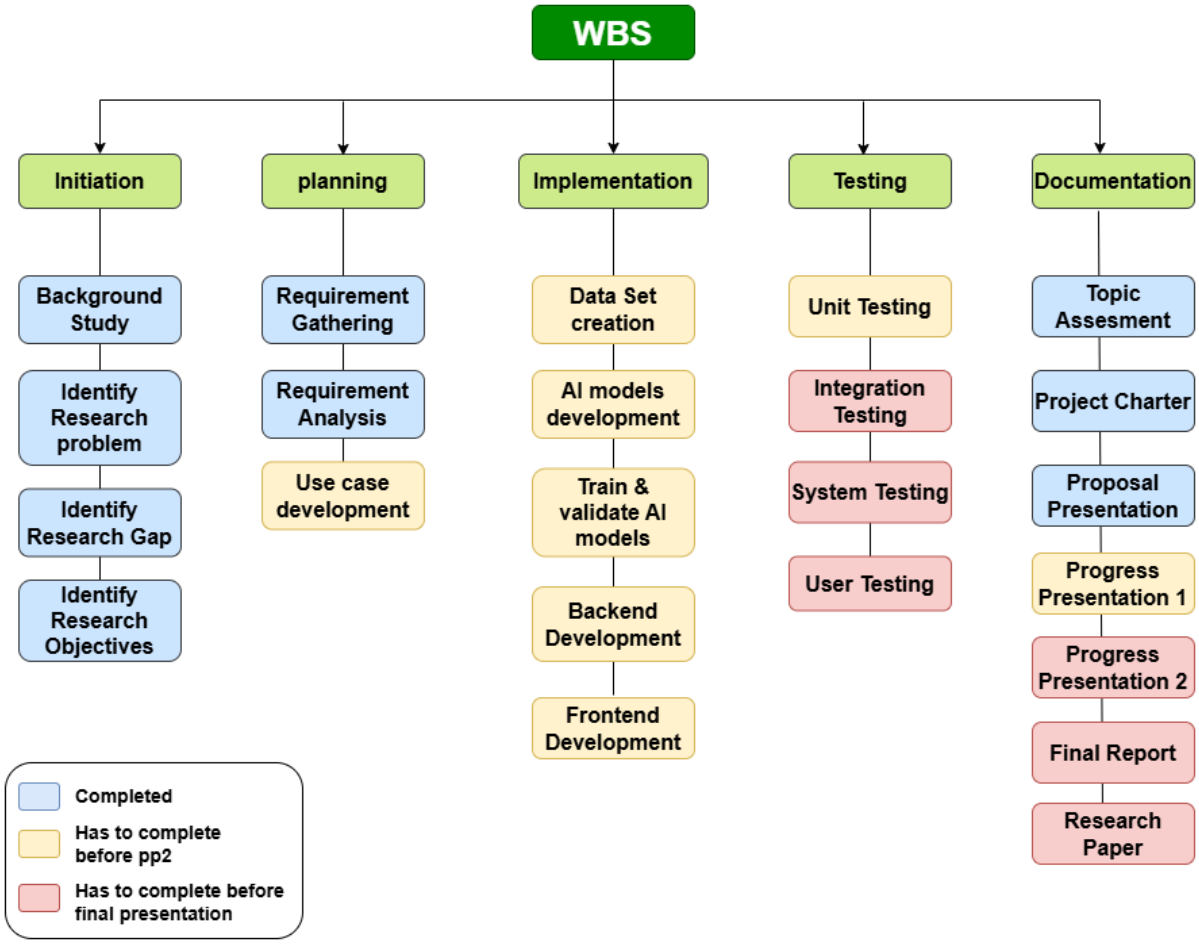


Figure 8.4: Work Break Down Structure

9. GANTT CHART

Table 9.1: GANTT CHART

	Semester 1					Semester 2						
Activity	1	2	3	4	5	6	7	8	9	10	11	12
Preparation	■											
Identify the Problem	■											
Requirement gathering	■											
Literature review		■										
Analyze requirements		■										
Online project registration		■										
Topic assessment from submission		■	■									
Identify main objectives		■	■									
Project proposal submission			■	■	■							
Data Collection					■							
Model Training					■	■	■	■	■	■	■	
Software Development							■	■	■	■	■	
Progress presentation 1							■	■	■			
Integration									■			
Testing									■	■	■	
Progress presentation 2										■		
Research paper submission										■	■	
Progress presentation 2										■	■	
Finalizing												■

Milestone Summary

Table 9.2: Milestone Summary

Milestone	**Months**
Hardware Ready	3
Data Collection Complete	4
Model Training Complete	3
Software Development Complete	3
Integration Complete	2
Testing Complete	2
Project Submitted	2

10. REFERENCES LIST

[1] A. Sungehetha, R. S. Sharma, S. B. P, S. Mahapatra, R. Thakur, and K. RC, "Multi-modal optical sensing fusion for quantitative monitoring of orchid parameters: Analysis of commercial cultivation in Asia," in 2025 2nd International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS), 2025.

[2] H. Hadi, S. N. S. Razali, and A. I. Awadh, “A Comprehensive Review of the Cosmeceutical Benefits of Vanda Species (Orchidaceae),” *Faculty of Pharmacy, International Islamic University Malaysia*, Kuantan, Malaysia, 2015.

[3] A. Setiaji, R. R. R. Annisa, A. Kinasih, A. D. Santoso, and A. D. R. Riyadi, “In vitro propagation of Vanda orchid: a review,” Gadjah Mada University, Yogyakarta, Indonesia.

[4] B. Srisompun, S. Boonmee, and P. Srisompun, "Paphiopedilum orchid flower classification using color moments, color histogram and SFTA texture features with MLP neural network," in Proceedings of IEEE Conference, 2024, pp. 1-7.

[5] L. B. Chen, G. Z. Huang, X. R. Huang, and W. C. Wang, "A self-supervised learning-based intelligent greenhouse orchid growth inspection system for precision agriculture," IEEE Sensors Journal, vol. 22, no. 24, pp. 1-11, Dec. 2022.

[6] K. Chand, S. Shah, J. Sharma, M. R. Paudel, and B. Pant, “Isolation, characterization, and plant growth promoting activities of endophytic fungi from a wild orchid Vanda cristata.”

[7] S. C. Yuan, S. Lekawatana, T. D. Amore, F. C. Chen, S. W. Chin, D. M. Vega, and Y. T. Wang, "The global orchid market," in *The Orchid Genome*, 2021, pp. 1-28.

[8] K. P. S. Chauhan and A. K. Singh, "Orchid cultivation: A growing business opportunity for farmers," *Journal of Horticultural Sciences*, vol. 14, no. 2, pp. 89-95, 2019.

[9] X. Wang, M. Liu, and J. Zhang, "Intelligent orchard technologies: A review of current status and future prospects," *Computers and Electronics in Agriculture*, vol. 185, p. 106125, 2021.

[10] R. P. Medhi and S. Chakrabarti, "Economics of orchid cultivation in North East India," *Indian Journal of Agricultural Economics*, vol. 72, no. 3, pp. 412-420, 2017.

[11] S. Ketsa and W. G. van Doorn, "Postharvest physiology and handling of orchid flowers," *Chronica Horticulturae*, vol. 58, no. 3, pp. 16-23, 2018.

[12] T. T. H. Nguyen and D. T. Nguyen, "Economic efficiency of high-tech orchid farming in Vietnam," *Journal of Asian Business and Economic Studies*, vol. 28, no. 4, pp. 245-260, 2021.

[13] M. S. Hadi, S. B. S. Rizki, M. A. As-Shidiqi, M. L. Arrohman, D. Lestari, and M. Irvan, "Smart greenhouse control system for orchid growing media based on IoT and fuzzy logic technology," in *Proc. 1st Int. Conf. Electron. Electr. Eng. Intell. Syst. (ICE3IS)*, 2021, pp. 165-169.

[14] A. K. Singh and R. K. Singh, "Low-cost IoT solutions for small and marginal farmers in developing countries," *Journal of Agricultural Informatics*, vol. 11, no. 2, pp. 45-56, 2020.

11. APPENDICES

Appendix A: Farmer Interview Questions

Topic: Understanding how farmers monitor Vanda orchid growth

How many Vanda orchids do you grow?
How do you currently monitor plant growth?
How do you know when the flowers will bloom?
How many days in advance can you now predict flowering?
Have you ever lost money because flowers bloom early or late?
What problems do you face with your current system?
Would you like a system that tells you the growth stage and predicts flowering 21 days in advance?
How much do you have to pay monthly?
What features are most important to you?

Appendix B: Model Flow Diagrams

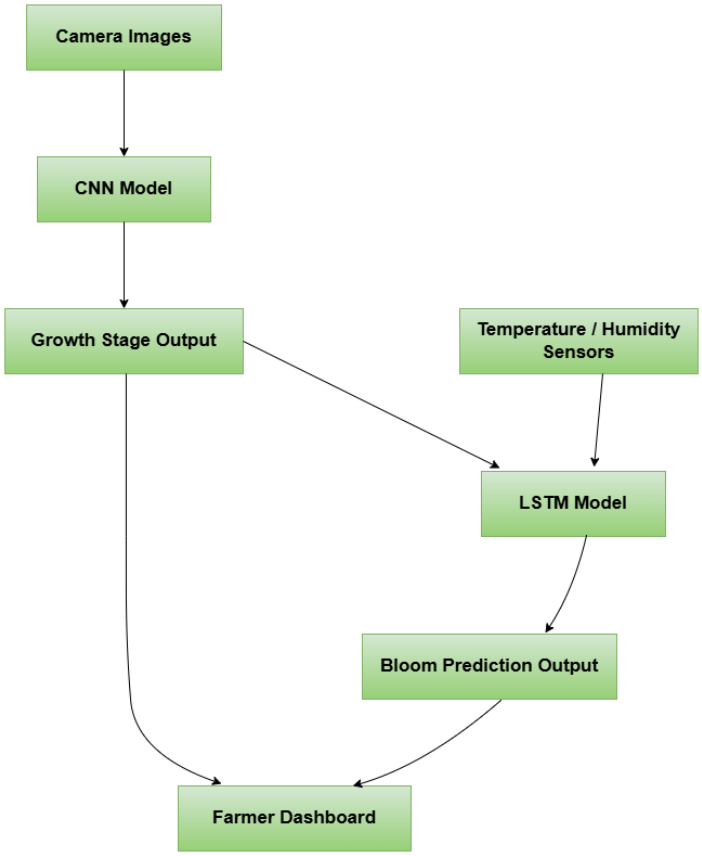


Figure 11.1: model Flow Diagram

Appendix C: UI Mockups

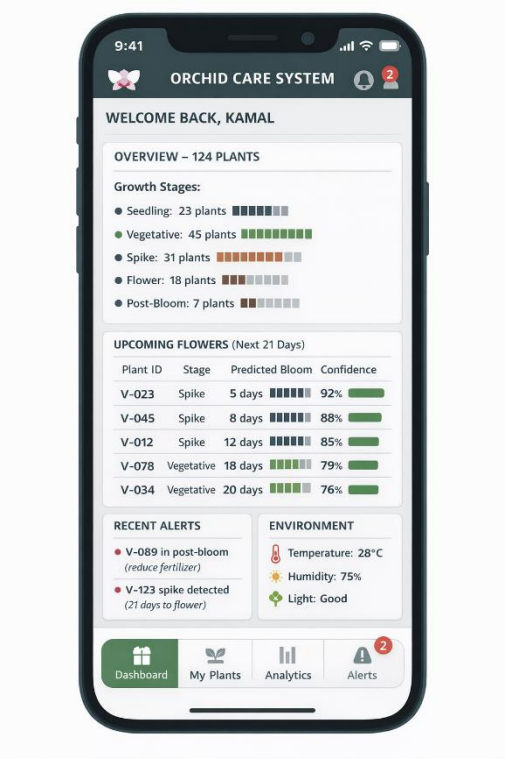


Figure 11.2: Farmer Dashboard - Main View

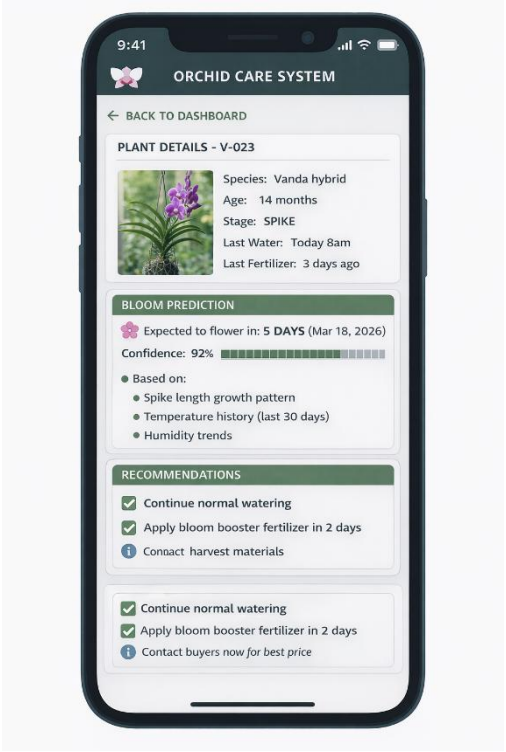


Figure 11.3: Plant Detail View

Appendix D: Risk Analysis

Table 11.1: Technical Risk Analysis

Risk Category	Risk Description	Likelihood	Impact	Mitigation Strategy
Technical Risks	Model accuracy below 90%	Medium	High	Start with simpler CNN architecture; use transfer learning from existing models [4]; collect more training data if needed
	Hardware failure (camera, sensors)	Low	Medium	Keep spare components; test thoroughly before deployment
	Data loss during transmission	Low	Medium	Local backup on SD card; daily cloud sync
	Poor image quality due to lighting	Medium	Medium	Install at fixed time daily; use consistent artificial lighting
Data Collection Risks	Insufficient images collected	Low	High	Start collection early (Month 3); capture multiple angles if needed
	Incorrect labeling of growth stages	Medium	High	Consult orchid experts for verification; cross-check labels
	Greenhouse access denied	Low	High	Get written permission early; have backup greenhouse location
Project Timeline Risks	Hardware delivery delayed	Low	Medium	Order components early (Month 1-2); use local suppliers if possible
	Model training takes longer	Medium	Medium	Use Google Colab for faster processing; parallelize work
	Farmer unavailable for feedback	Low	Low	Schedule visits in advance; offer small incentive
Deployment Risks	Farmers find system difficult to use	Medium	Medium	Simple dashboard design; provide training session; get feedback early
	System not affordable for target users	Medium	Medium	Use low-cost hardware (LKR 17,000 per unit); subscription pricing kept low [14]
Security Risks	Unauthorized access to data	Low	High	Password protection; encrypted transmission; role-based access
	Physical theft of hardware	Low	Medium	Locked enclosures; install in secure location